Harsh Sinha

Cover Letter

Software Engineer, Intern

Hi,

I'm Harsh, a pre-final year Integrated MTech student in Computer Science at IIIT Bangalore, applying for the Software Engineer Intern role.

I wanted to apply here because I like the idea of interning somewhere my work actually goes live, not just something I build and then it sits in a repo. Last year I built Food-e, a mess management system for my college. I did the backend on my own, login and OAuth, role based access for four types of users, caching, PDF reports. Right now it's used by 1,700+ students, with around 900 opening it every day. Once people started actually using it daily, I had to get better about fixing things fast. If something breaks, someone tells me almost immediately, and I fix it that day.

I also did a summer internship at IIIT Bangalore's Centre for Technology Research, where I worked on an existing citizen complaint portal instead of something I built from scratch. I had to spend time understanding how it was built before I could add new APIs or set up OAuth and role based access on top of it. That felt like a good exercise in reading someone else's code properly instead of just writing my own.

More recently I've been working on a small project with a job queue and a few worker processes running in parallel. I ran into some concurrency problems along the way, threads competing for resources, database connections running out, that kind of thing. I fixed them one at a time and got the throughput up by 1.6x. It took a while to figure out but I learned a fair bit doing it.

Outside of college work I play guitar sometimes, mostly just to take a break from screens.

I think I'd be a good fit here because I like taking on a problem I don't fully understand yet, spending time with it, and being responsible for it end to end. Being a TA for over 120 students also taught me to explain things clearly instead of assuming people will just get it.

Thanks for reading this.

Harsh